

MGX™ SuperServer ARS-121L-DNR-LCC

1U 2-Node NVIDIA Grace™ CPU Superchip liquid-cooled system supporting NVIDIA BlueField®-3 or NVIDIA ConnectX®-7 (complete rack and onsite service are required)



Key Applications

High Performance Computing, Hyperscale Cloud Applications, Data Analytics,

Key Features

- Two **hot-pluggable** systems (nodes) in a 1U form factor. Each node supports the following;;
- High density processor(s) NVIDIA 144-core CPUs on a Grace™ CPU Superchip with liquid- cooling;
- Up to 960GB onboard memory;
- Up to 2xPCIe 5.0 x16 slots. Please note: GPU is not supported;
- Up to 2xE1.S NVMe drive bays, 2xM.2 NVMe drive;
- 2 Redundant 2000W Titanium Level power supplies;



Form Factor	1U Rackmount Enclosure: 438.4 x 43.6 x 900mm (17.26" x 1.7" x 35.4") Package: 695 x 230 x 1170mm (27.36" x 9.05" x 46.06")
Processor	Single processor(s) NVIDIA Dual 72-core CPUs on a Grace™ CPU Superchip
System Memory	Onboard Memory Up to 960GB ECC
Drive Bays Configuration	Default: Total 2 bays • 2 front fixed E1.S NVMe drive bays M.2: 2 M.2 NVMe slots (M-key)
Expansion Slots	Default • 1 PCIe 5.0 x16 FHFL slot • 1 PCIe 5.0 x16 LP slot
On-Board Devices	NVIDIA C2
Input / Output	LAN: 1 RJ45 1 GbE Dedicated BMC LAN port USB: 1 port(Rear) Video: 1 mini-DP port

(Front View – System)



Service/Asset Tag with BMC Password

Control Panel

Drive Bay	Description
1 2	2 Fixed E1.S NVMe Drive Bays

(Rear View – System)



Mini DisplayPort

2 Redundant 2000W Titanium Level Power Supplies

USB Port

LAN Port

System Cooling	Fans: Up to 7 Removable heavy-duty 4cm Fan(s) Liquid Cooling: Direct to Chip (D2C) Cold Plate
Power Supply	2x 2000W Redundant Titanium Level (96%) power supplies
System BIOS	BIOS Type: AMI 64MB SPI Flash EEPROM
PC Health Monitoring	CPU: Monitors for CPU Cores, Chipset Voltages, Memory FAN: Fans with tachometer monitoring Pulse Width Modulated (PWM) fan connectors Status monitor for speed control Temperature: Monitoring for CPU and chassis environment Thermal Control for fan connectors
Dimensions and Weight	Weight: Gross Weight: 49.21 lbs (22.32 kg) Net Weight: 33.64 lbs (15.26 kg) Available Color: Silver
Operating Environment	Operating Temperature: 10°C to 35°C (50°F to 95°F) Non-operating Temperature: -40°C to 60°C (-40°F to 140°F) Operating Relative Humidity: 8% to 90% (non-condensing) Non-operating Relative Humidity: 5% to 95% (non-condensing)
Motherboard	Super G1SMH
Chassis	CSE-MG102TS-R000NDFP-2N